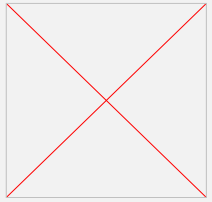
Lab 4-5
Operational Amplifier

Electronics and Photonics Post-undergraduate Second Degree Program

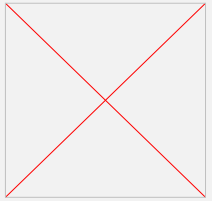
National Yang Ming Chiao Tung University

NYCU 學士後電子與光子學士學位學程

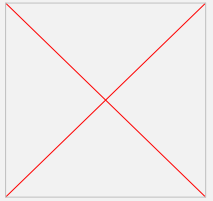
Outline



Noninverting Amplifier & Inverting Amplifier



Summing Amplifier & Subtracting Amplifier



Integrator & Differentiator

Noninverting Amplifier & Inverting Amplifier

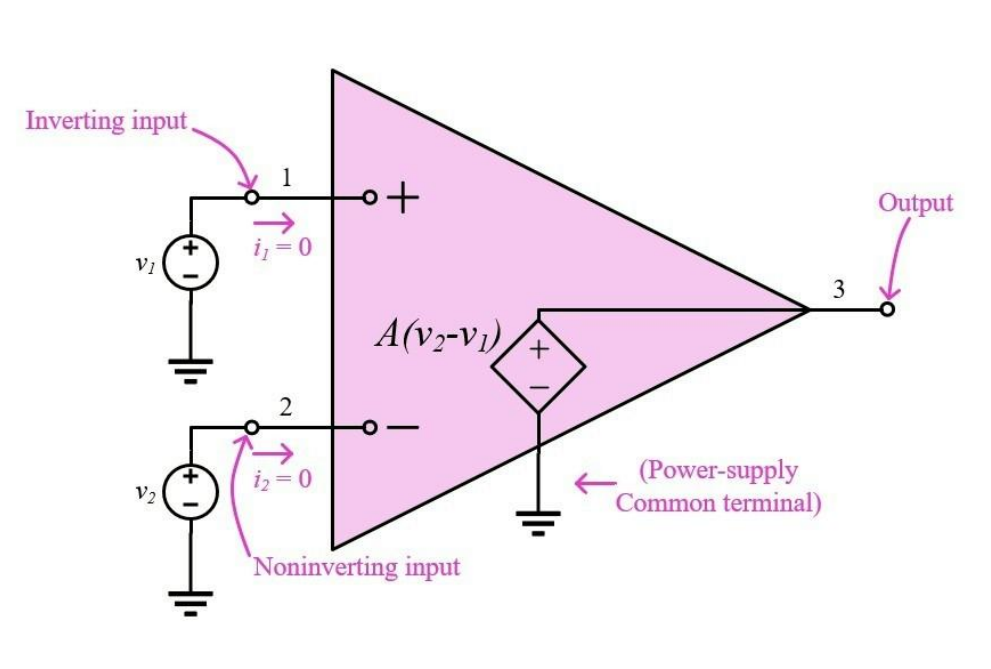
Theory overview

Ideal model for op-amp must satisfied this five condition

- Infinite input impedance
- Zero output impedance
- Zero common-mode gain or, equivalently, infinite common-mode rejection
- Infinite open-loop gain
- Infinite bandwidth

Theory overview

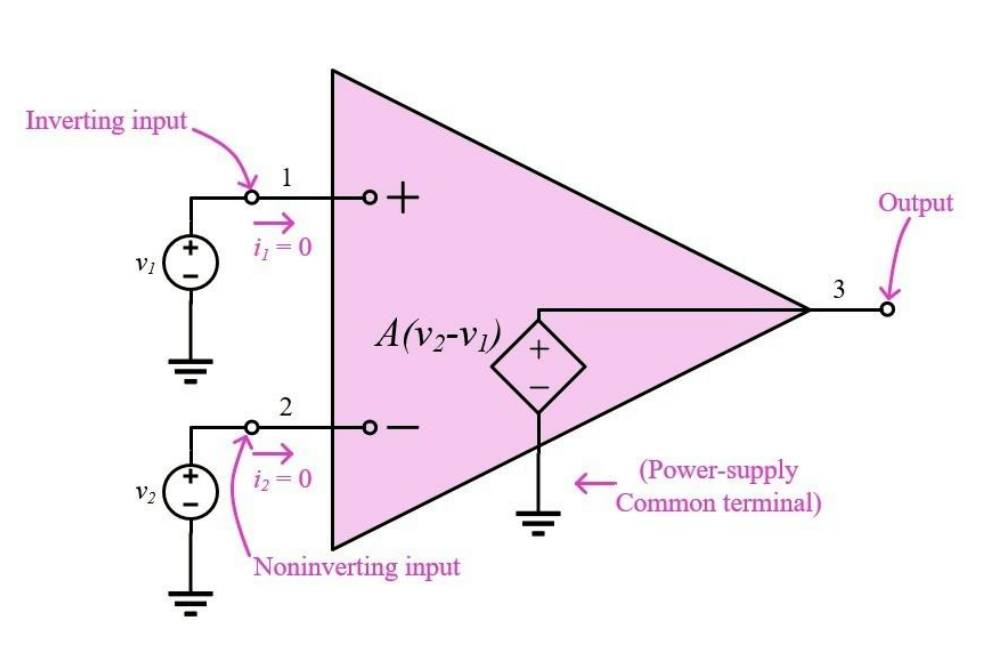
Ideal model for op-amp



- Linear Mode : the output through negative feedback pull to the input terminal, in this case, $V_{in} = V_+ = V_-$,it calls **virtual short**.
- the input impedance is infinite, resulting in no current flowing into

Theory overview

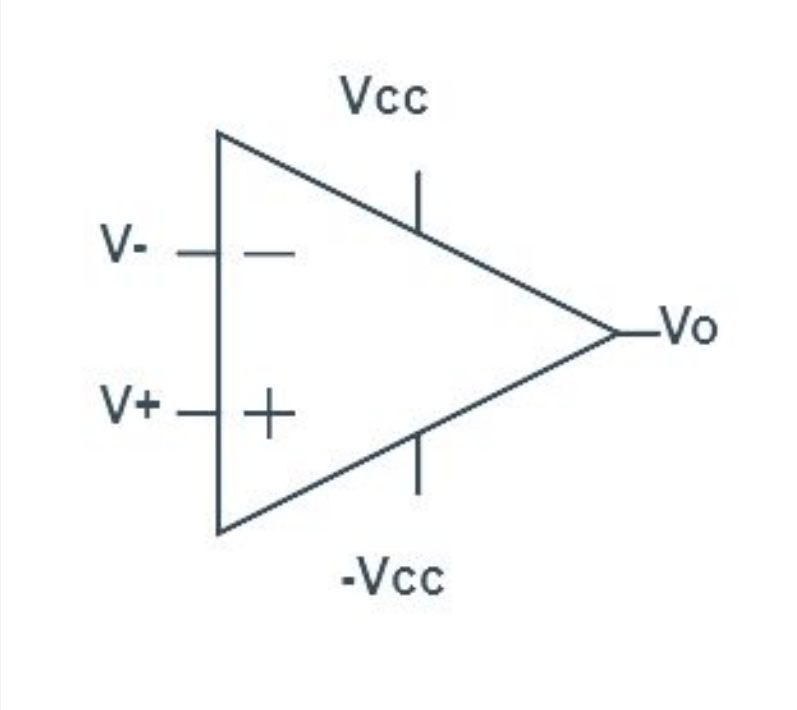
Non-ideal model for op-amp



- the input impedance is not considered infinite.
- Must consider input current
- the virtual short between the input terminals becomes invalid.
- the output voltage:

$$V_O = A_V \times (V_+ - V_-)$$

Theory overview



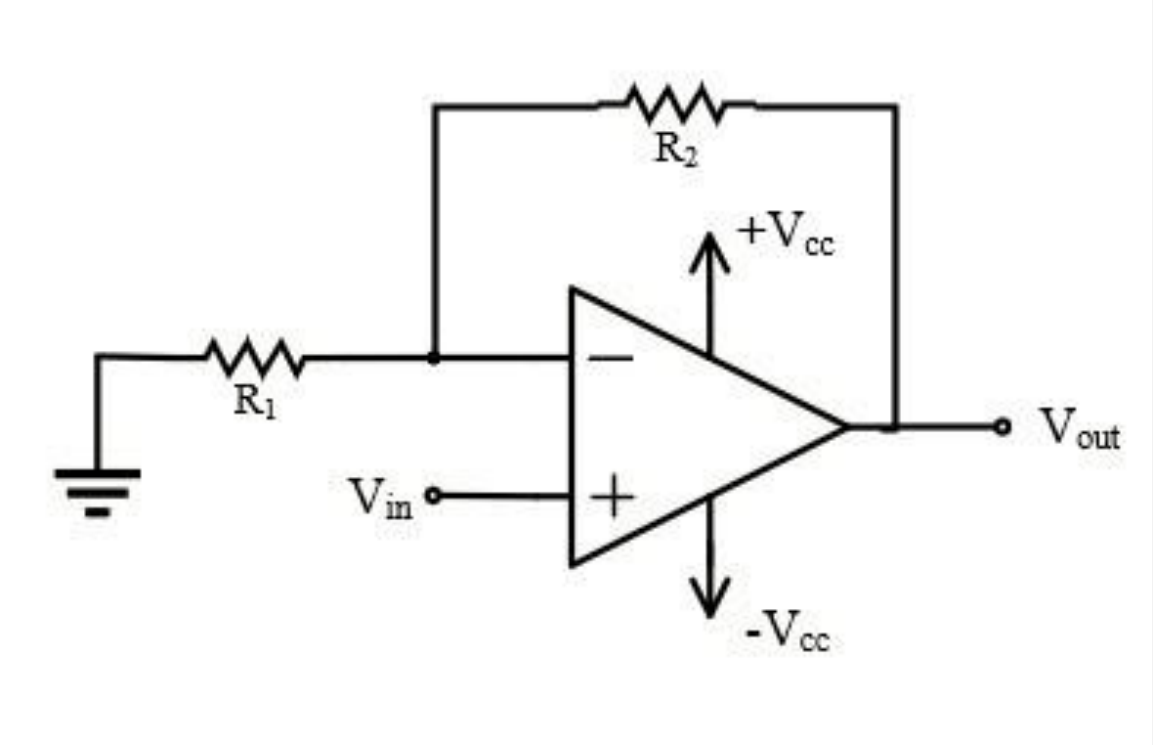
Terminal Name	Symbol	Description
Non-inverting input		Input that does not invert the signal
Inverting input		Input that inverts the signal
Output		Amplified signal output
Positive power supply		The maxima level for the output signal
Negative power supply		The minima level for the output signal

Theory overview

Non-inverting amplifier (satisfied ideal case)

- R_2 cause negative feedback, the input signal input by inverting terminal

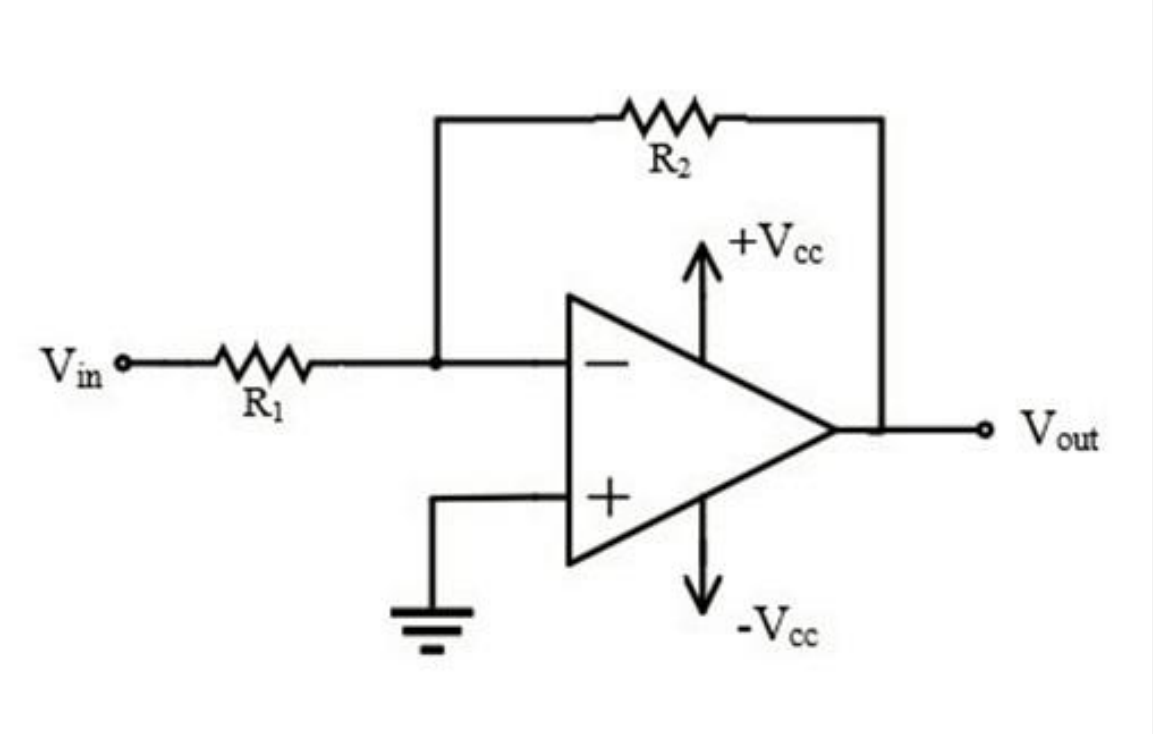
- $V_- = V_O + \frac{R1}{R_1+R_2}$
- $V_{in} = V_O + \frac{R1}{R_1+R_2}$
- $A_V = \frac{V_o}{V_{in}} = 1 + \frac{R_2}{R_1}$



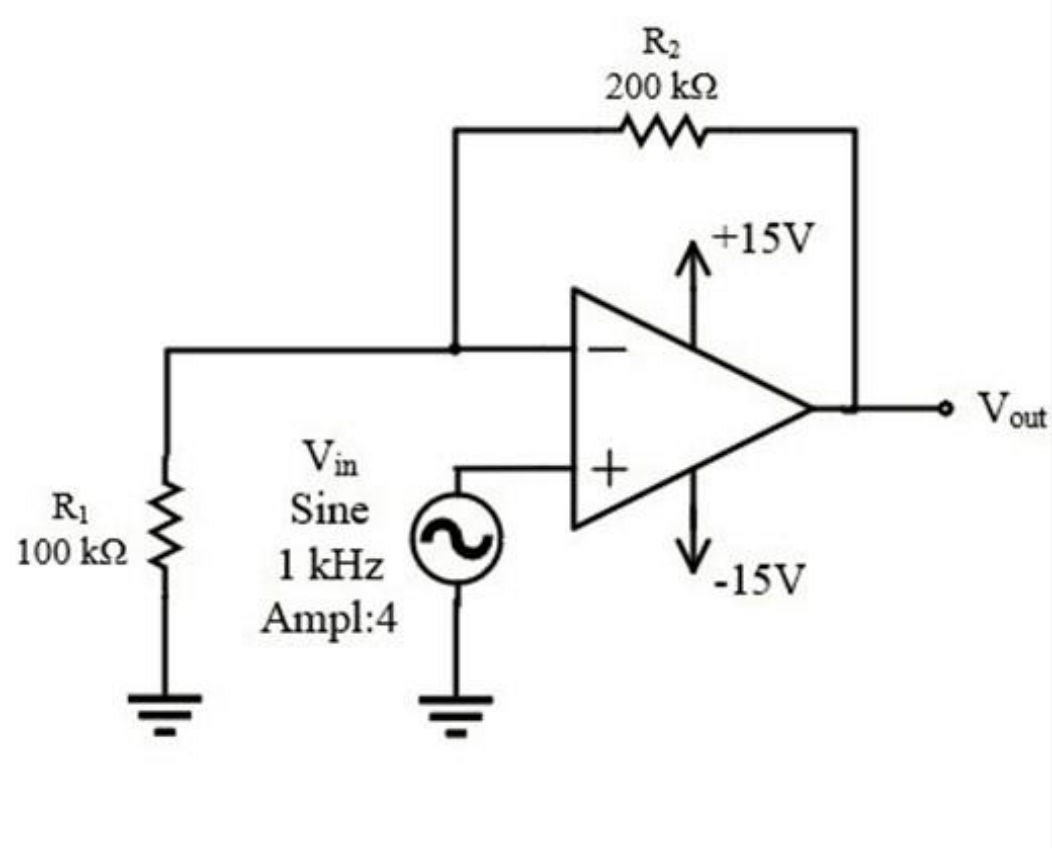
Theory overview

inverting amplifier (satisfied non-ideal case)

- R_2 cause negative feedback, the input signal input by non-inverting terminal
- $\frac{V_{in}-0}{R_1} = \frac{0-V_o}{R_2}$
- $A_V = \frac{V_o}{V_{in}} = -\frac{R_2}{R_1}$



Circuit schematic



1. Build the circuit and using V_{in} and also supply two DC voltage sources.

2. Measure the output voltage and waveform of the circuit.

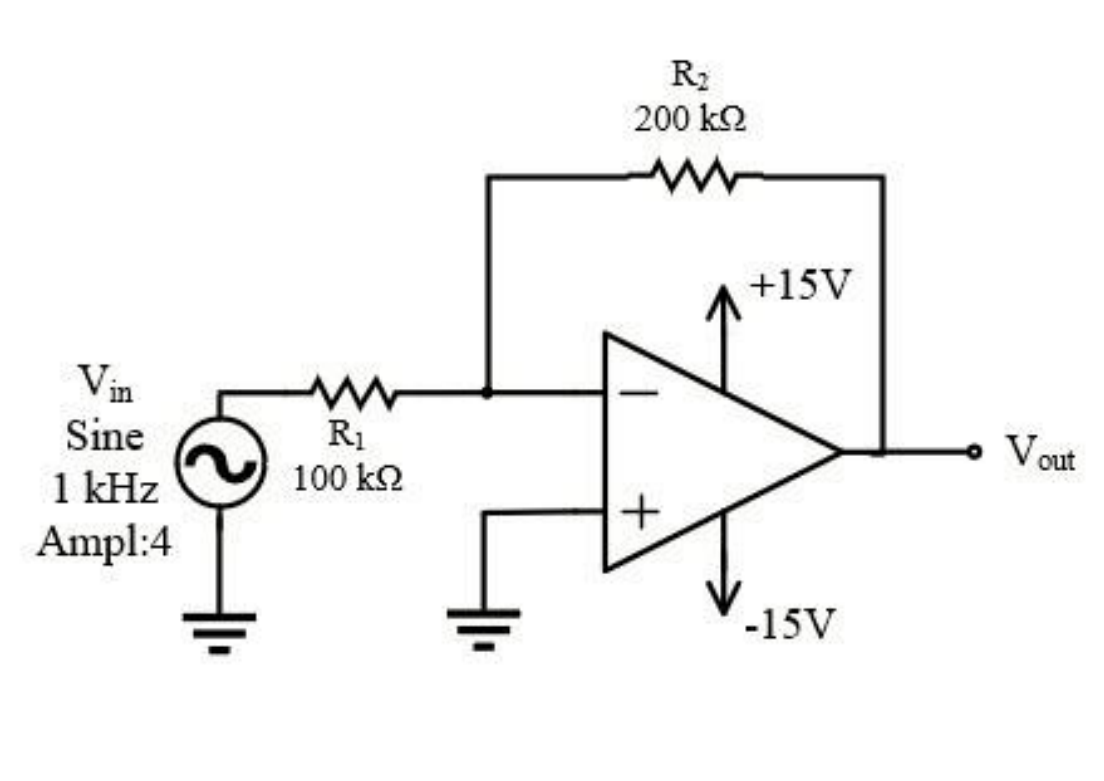
3. Calculate the voltage gain (A_v) of the circuit configuration.

- $A_V = \frac{V_o}{V_{in}} = 1 + \frac{R_2}{R_1}$ v_o = _____

	Input waveform	Output waveform
Waveform		

A_v (theoretical calculation)	A_v (experimental calculation)

Circuit schematic



1. Build the circuit and using V_{in} and also supply two DC voltage sources.
2. Measure the output voltage and waveform of the circuit.
3. Calculate the voltage gain (A_v) of the circuit configuration.

- $A_v = \frac{V_o}{V_{in}} = -\frac{R_2}{R_1}$ v_o = _____

	Input waveform	Output waveform
Waveform		

A_v (theoretical calculation)	A_v (experimental calculation)

Component list

Component	Specification	Number
Op-Amp	UA741	1
Resistor	100 kΩ (1/4 W)	1
	200 kΩ (1/4 W)	1
DC Power Supply	GW Instek GPE-3323	1
Signal Source (Function Generator)	GW Instek AFG-2225	1
Oscilloscope	Keysight EDUX1052G	1
Multimeter	GW Instek GDM-532	1

Question

1. What are the practical differences between inverting and non-inverting amplifiers?

2. Does a non-inverting amplifier have a higher or lower input impedance compared to an inverting amplifier, and what advantages does this characteristic offer in circuit design?

3. In Inverting Amplifier, if the input signal frequency is increased from 1 kHz to 1 MHz, will the circuit still provide accurate amplification?

Bonus

Design a non-inverting amplifier circuit with a voltage gain is equal to 1 (unity gain) and derive the gain in terms of open-loop gain A_o.

Summing Amplifier & Subtracting Amplifier

Objective

- Explore the concepts of inverting and noninverting amplifier.
- Verify the characterristics of summing and subtracting amplifiers through experiments.
- Understand how output voltage is influenced by the ratio of input voltages.

Theory overview

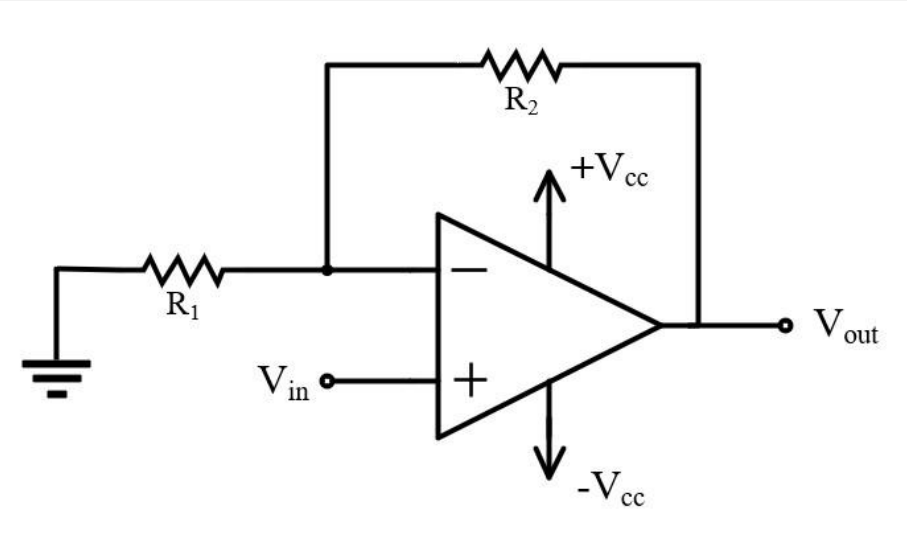
- In an ideal operational amplifier, gain is determined by external resistors.

- Use the superposition principle to analyze circuit with multiple inputs.

- Three main configurations:
  1. The inverting summing Amplifier
  2. The noninverting summing Amplifier
  3. The subtracting Amplifier

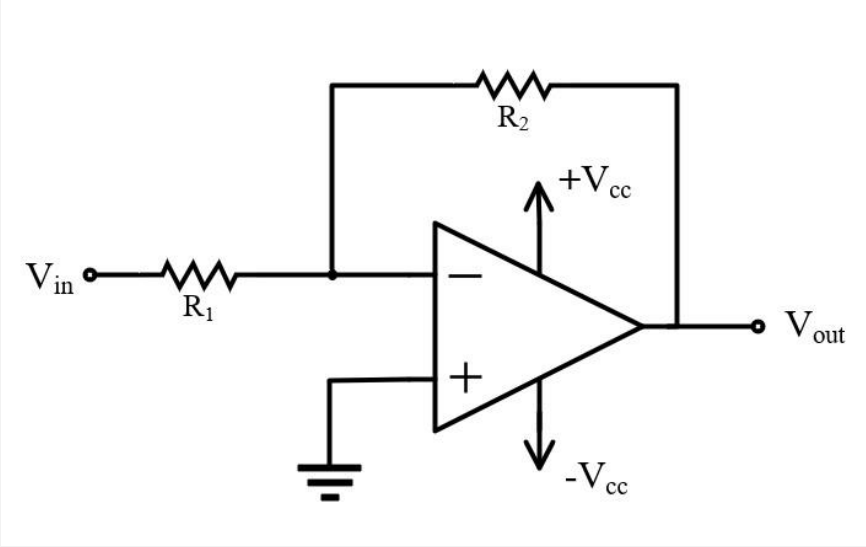
Theory overview

Non-inverting amplifier:



$$A_V = \frac{V_o}{V_{in}} = 1 + \frac{R_2}{R_1}$$

Inverting amplifier:



$$A_V = \frac{V_o}{V_{in}} = -\frac{R_2}{R_1}$$

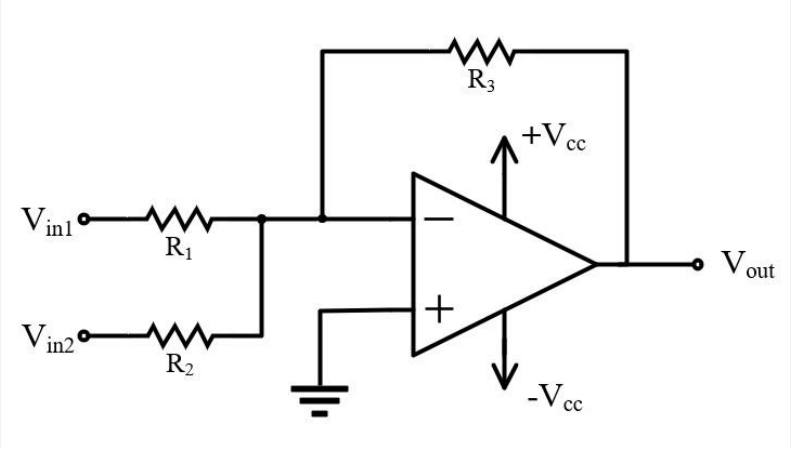
Theory overview

- **The inverting summing amplifier :**

$$V_{out} = \left(-\frac{R_3}{R_1}\right) \times V_{in1} + \left(-\frac{R_3}{R_2}\right) \times V_{in2}$$

→ if $R_1 = R_2 = R_3$

→ $V_{out} = -(V_{in1} + V_{in2})$

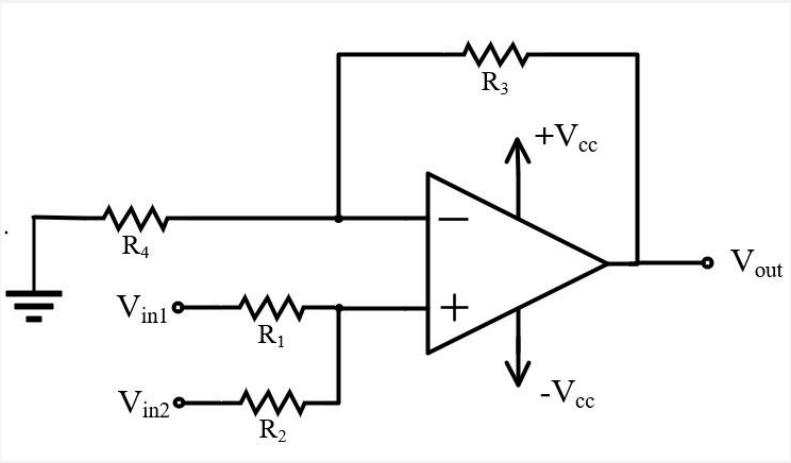


- **The summing amplifier :**

$$V_{out} = \left[\frac{R_2}{R_1 + R_2} \times \left(1 + \frac{R_3}{R_4}\right)\right] \times V_{in1} + \left[\frac{R_1}{R_1 + R_2} \times \left(1 + \frac{R_3}{R_4}\right)\right] \times V_{in2}$$

→ if $R_1 = R_2 = R_3 = R_4$

→ $V_{out} = V_{in1} + V_{in2}$



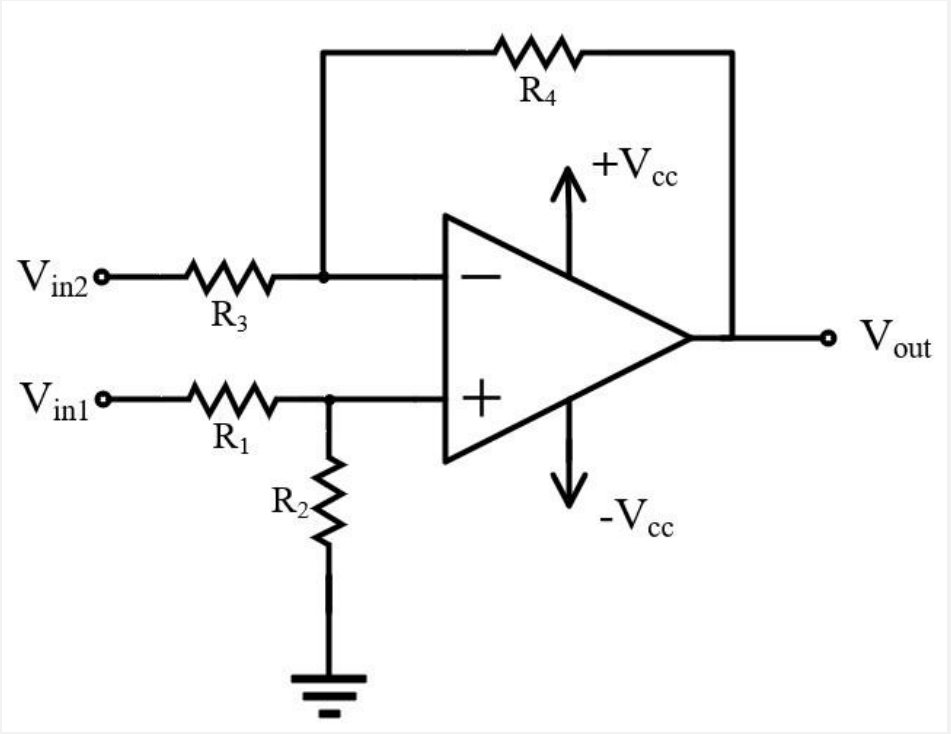
Theory overview

- **The subtracting Amplifier：**

$$V_{out} = \left[\frac{R_2}{R_1+R_2} \times \left(1 + \frac{R_3}{R_4}\right)\right] \times V_{in1} + \left(-\frac{R_4}{R_3}\right) \times V_{in2}$$

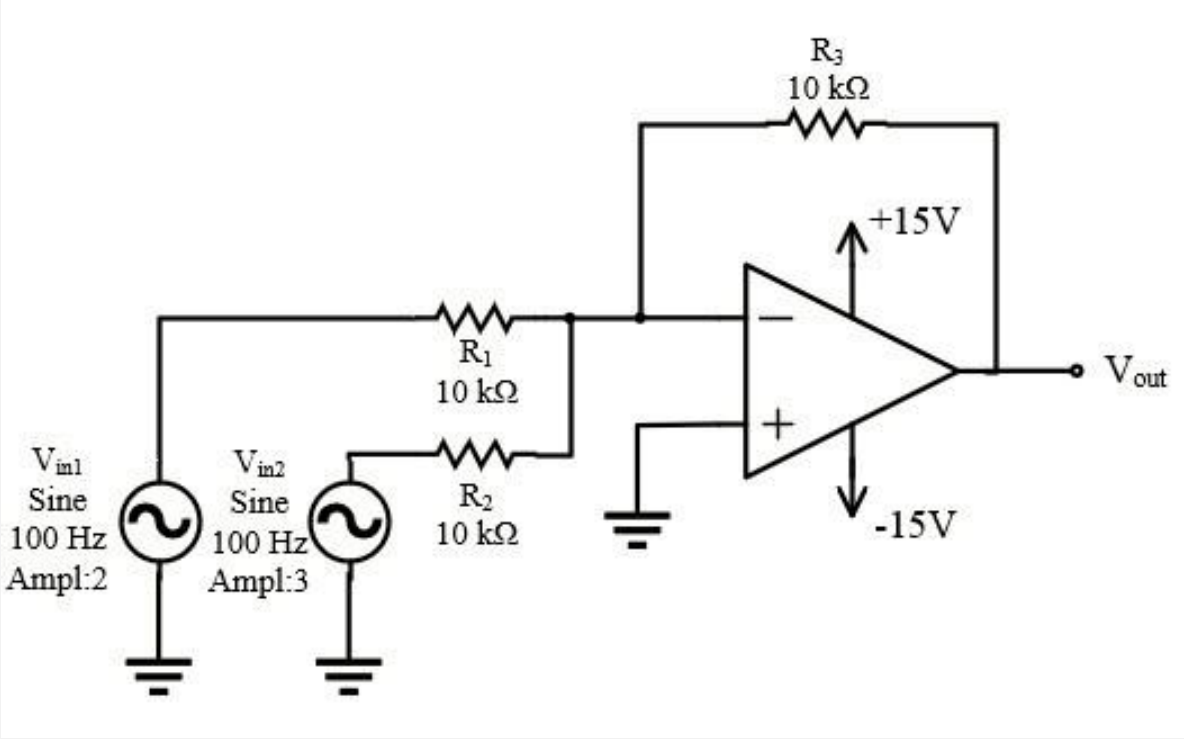
→ if $R_1 = R_2 = R_3 = R_4$

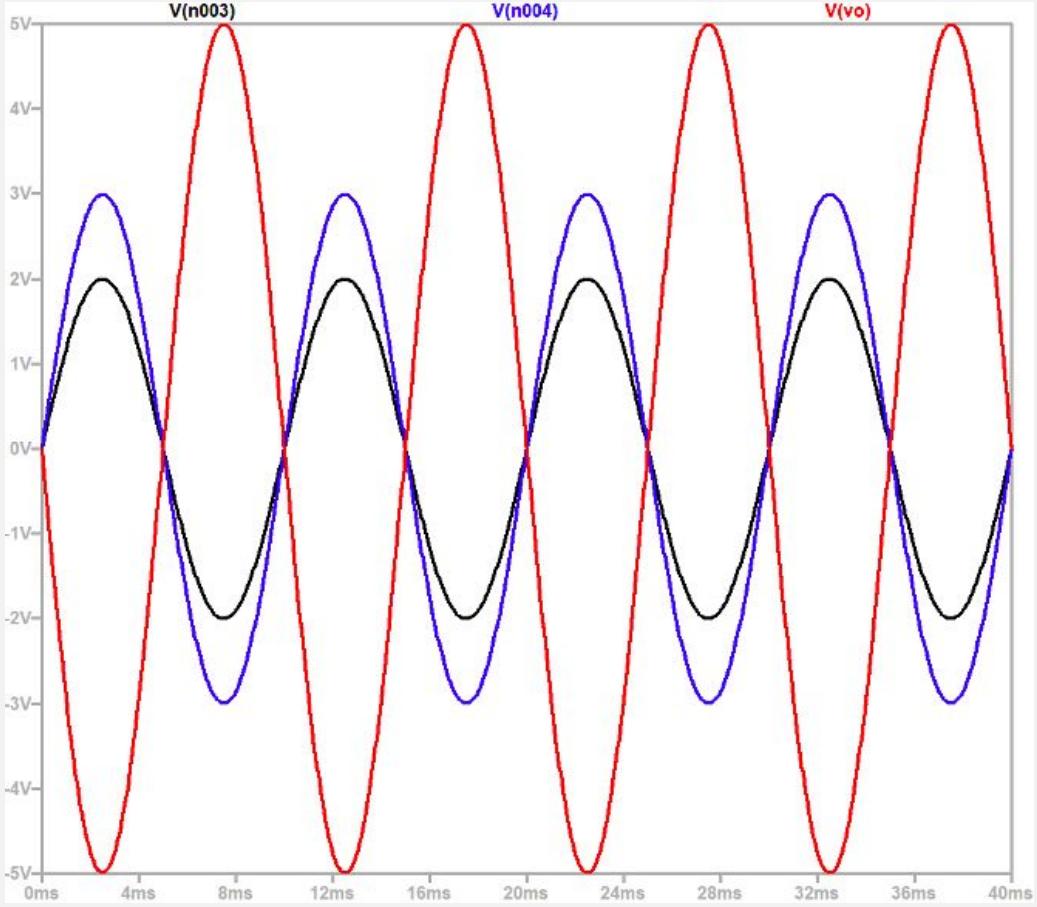
→ $V_{out} = V_{in1} - V_{in2}$



Circuit schematic

the inverting summing amplifier circuit configuration of an op-amp

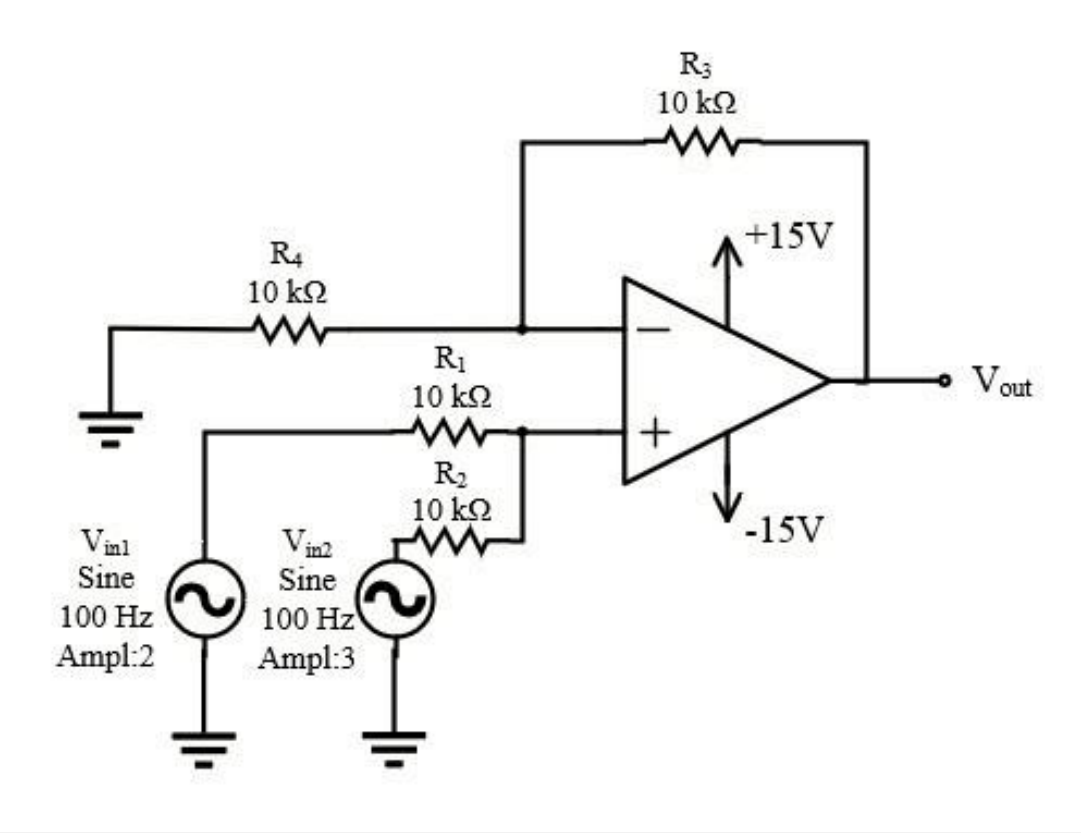


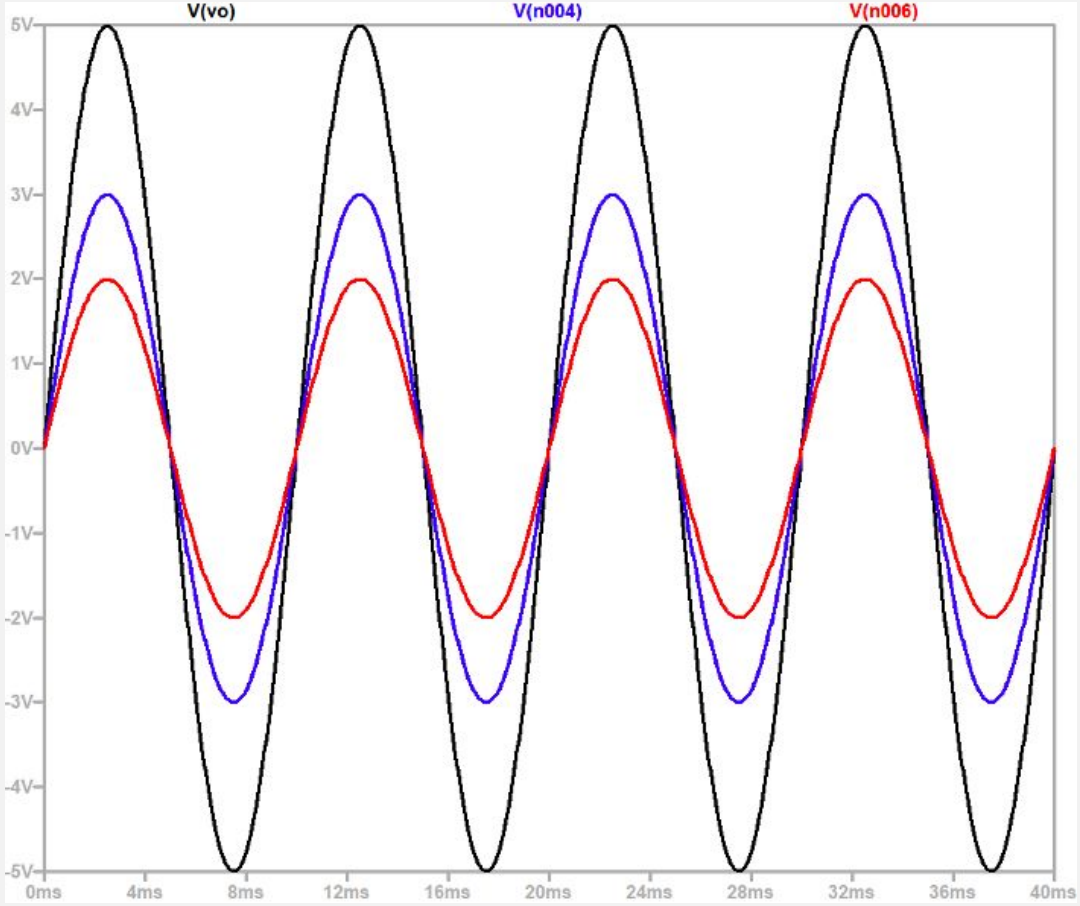


500 or 1k Hz 太低頻易出問題

Circuit schematic

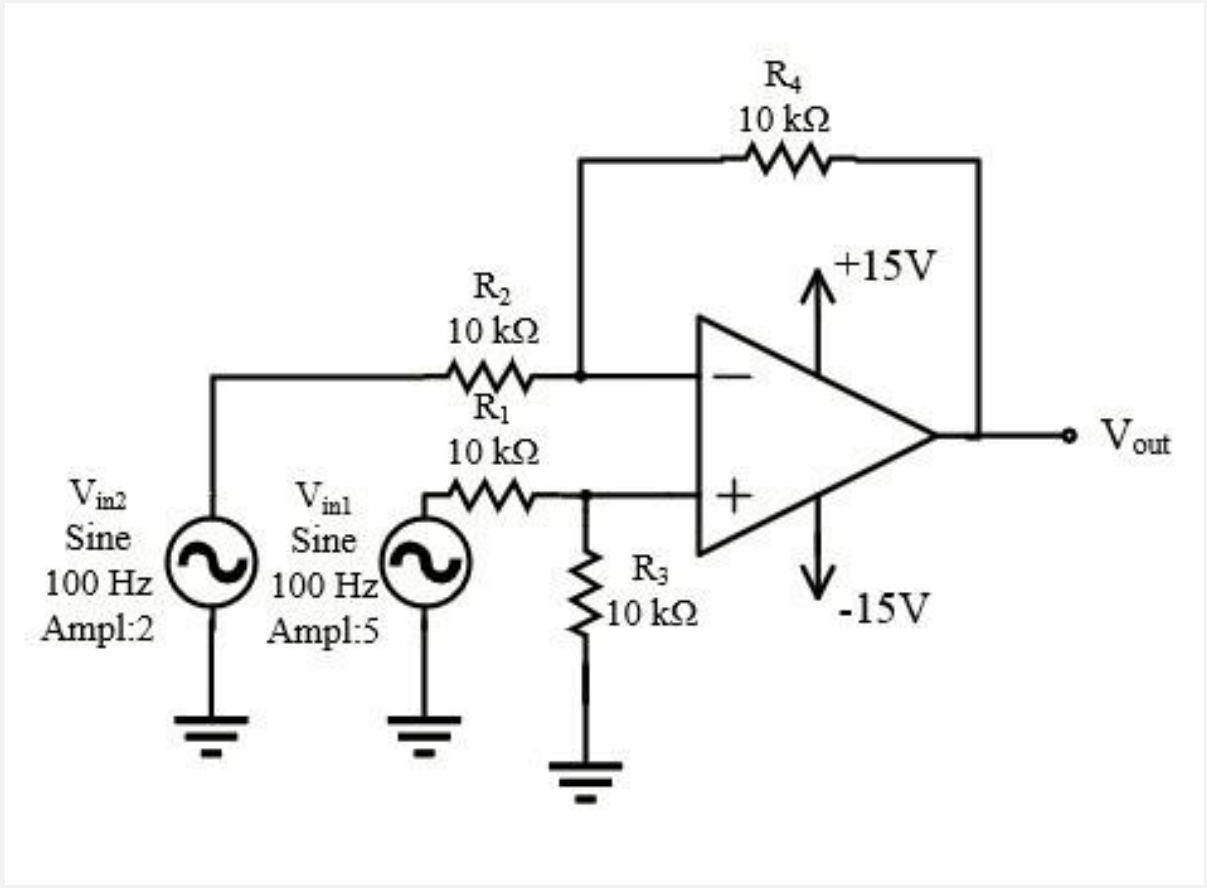
the summing amplifier circuit configuration of an op-amp

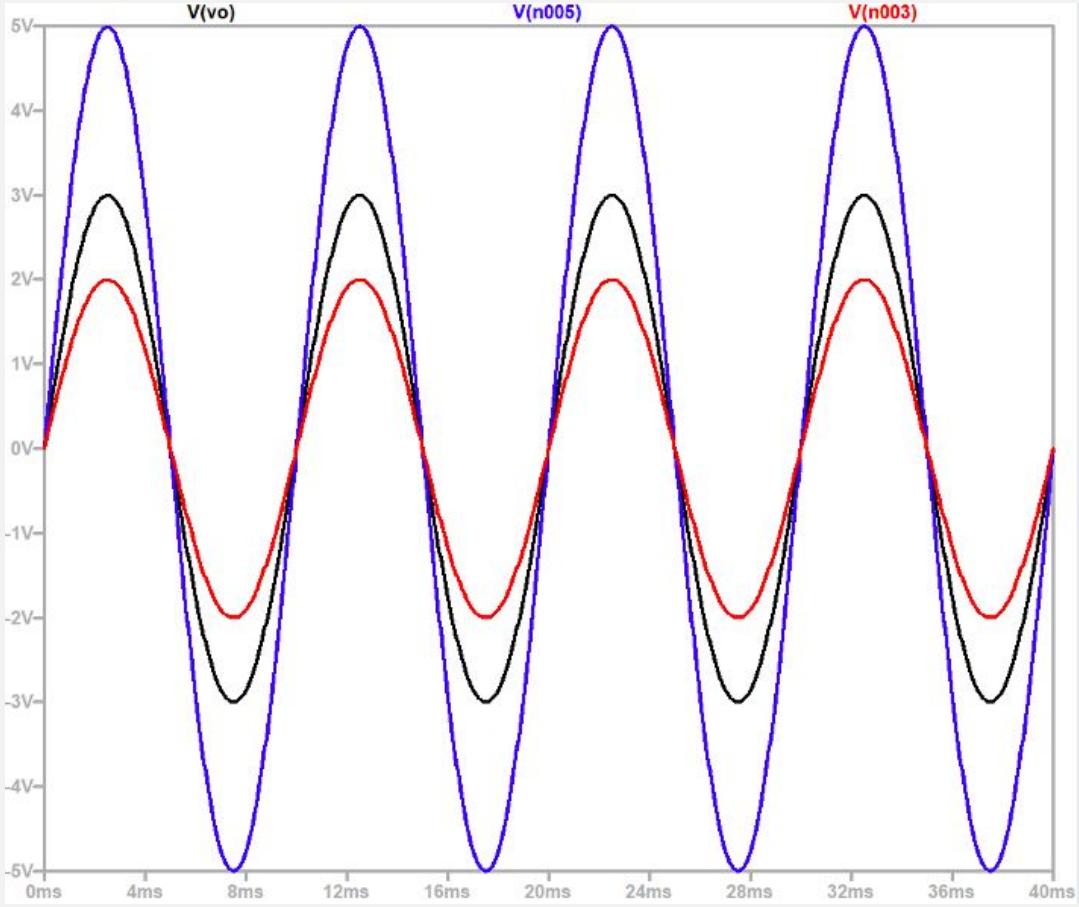




Circuit schematic

the subtracting amplifier circuit configuration of an op-amp





Experiment data table

	Input & output waveform
Waveform	

$V_{in1\text{-}pp}$ (the peak-to-peak voltage of input1)	$V_{in2\text{-}pp}$ (the peak-to-peak voltage of input2)	Output voltage (theoretical calculation)	Output voltage (experimental calculation)
V	V	V	V

Av (theoretical calculation)	Av (experimental calculation)

Component list

Component	Specification	Number
Op-Amp	UA741	1
Resistor	10 kΩ (1/4 W)	4
DC Power Supply	GW Instek GPE-3323	1
Signal Source (Function Generator)	GW Instek AFG-2225	1
Oscilloscope	Keysight EDUX1052G	1
Multimeter	GW Instek GDM-532	1

Question

1. In an ideal case, the subtracting amplifier should output 0 V when $V_{in1}=V_{in2}$. However, in the experiment, a small DC offset is observed at the output. List possible causes of this offset.

2. If you want the adder to have a weight of 3 for V_{in1} and a weight of 0.5 for V_{in2}, provide one possible resistor configuration (assume an inverting adder and that R_f can be freely chosen).

3. If you want to mix two audio signals (e.g., microphone and guitar inputs), which amplifier configuration would you choose — inverting or non-inverting? Please explain.

Bonus: Safely Mix Stereo to Mono

Your phone has a single speaker, but the audio source is stereo (**L** and **R**). Pick the coefficients $\boldsymbol{a}$ and $\boldsymbol{b}$ for

$$\boldsymbol{V_{mono} = -(aL + bR)}$$

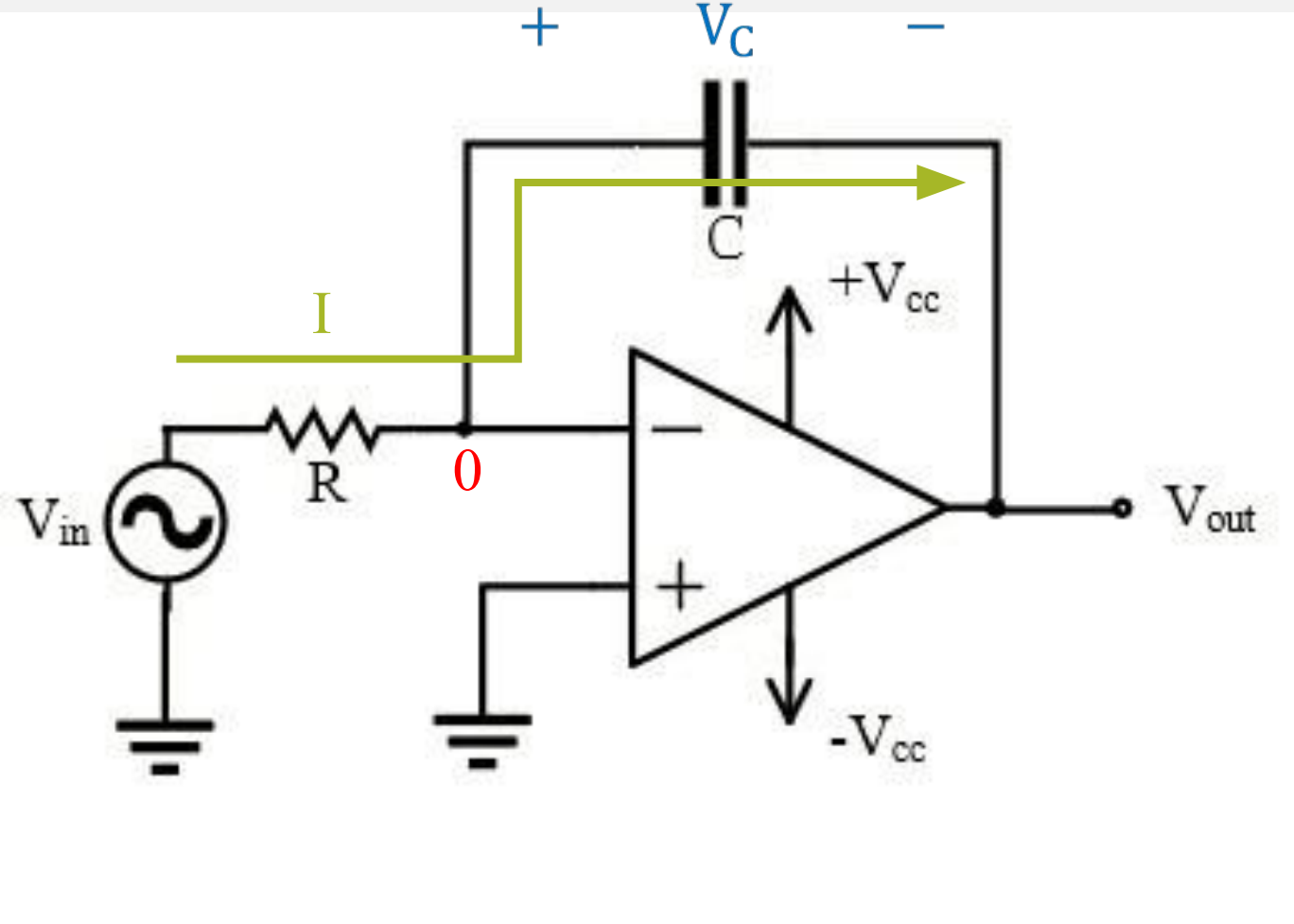
so that Left (L) and Right (R) are mixed to mono with equal weighting and the mono output never exceeds the amplitude of either input.

Integrator & Differentiator

Objective

- The objective of this experiment is to understand the basic operation of op-amp integrator and differentiator circuits, and to observe how the output waveforms represent the integration and differentiation of the input signals.

Theory overview

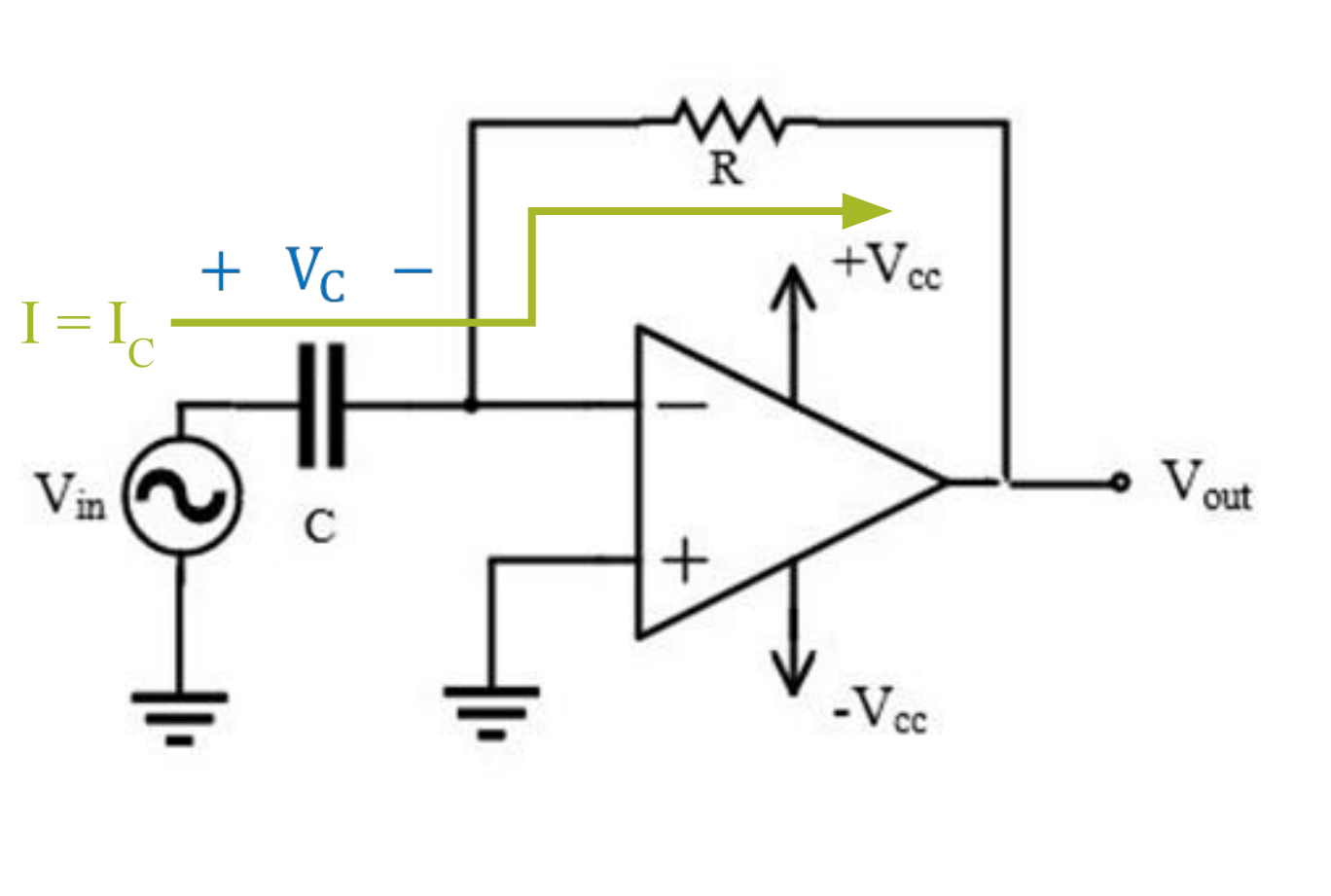


$$I = \frac{V_{in}}{R}$$

$$V_C = \frac{Q_C}{C} = \frac{1}{C}\int I_C\,dt = \frac{1}{C}\int \frac{V_{in}}{R}\,dt$$

$$V_{out} = -V_C = -\frac{1}{RC}\int_0^t V_{in}(t)\,dt$$

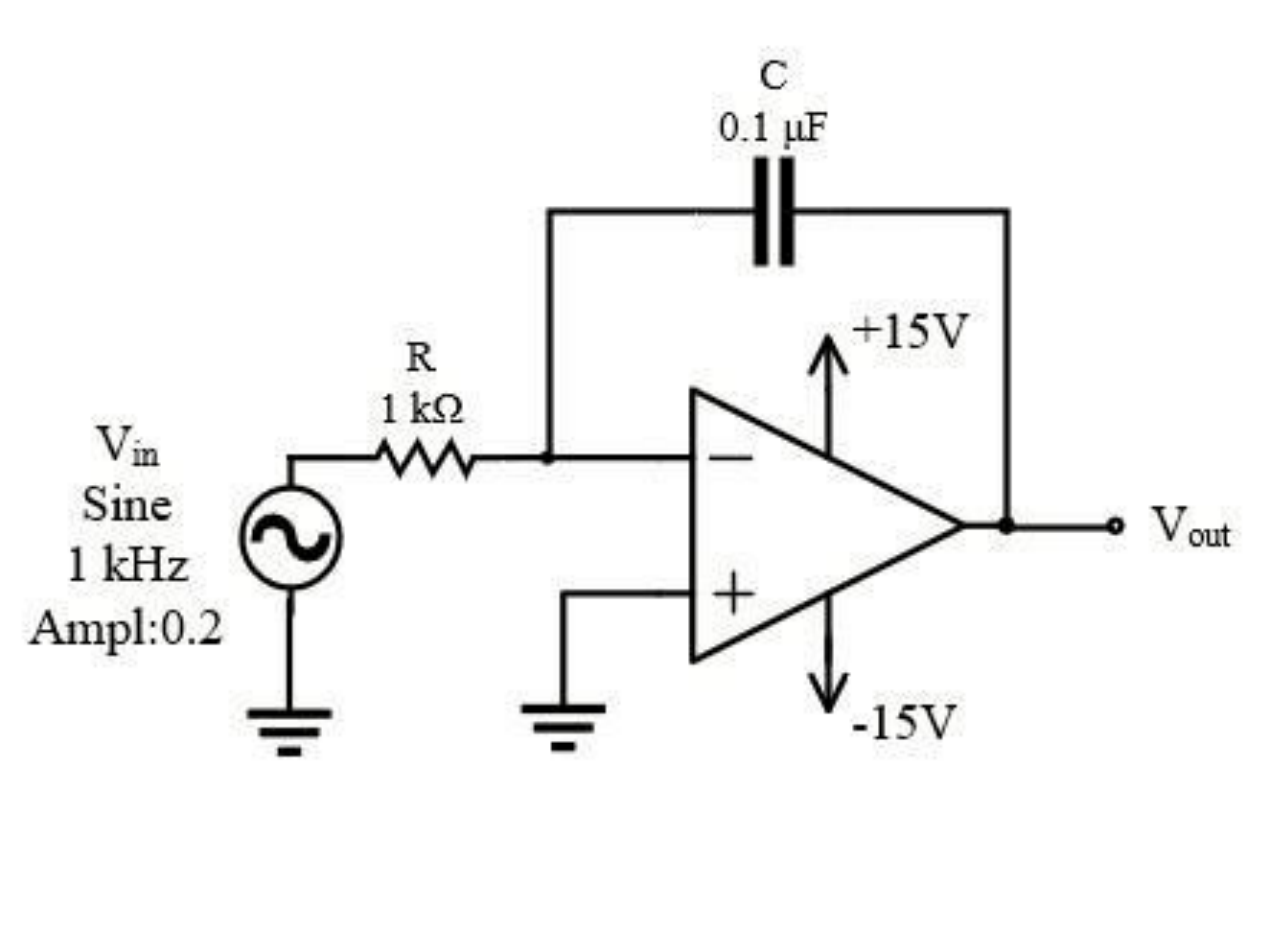
Theory overview

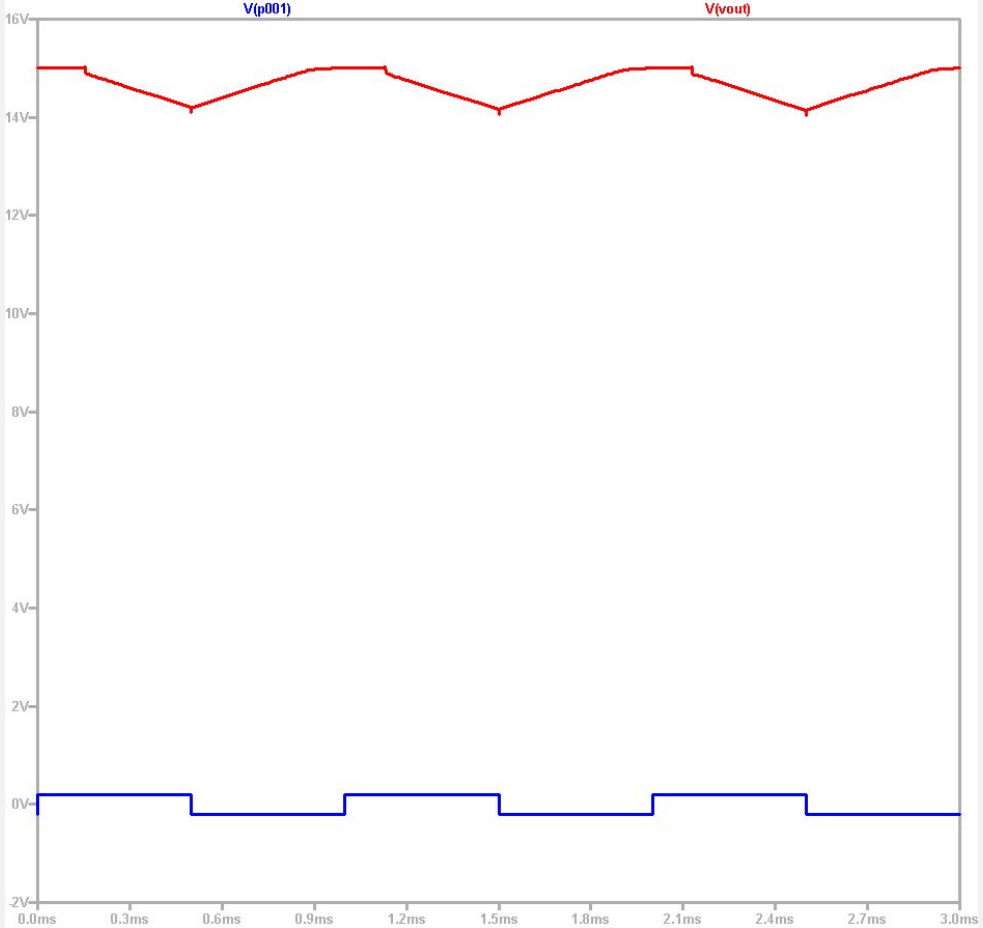


$$I_C = C\frac{dV_C}{dt} = C\frac{dV_{in}}{dt}$$

$$V_{out} = 0 - I \times R = -RC\frac{dV_{in}}{dt}$$

Circuit schematic





Experiment data table

	Input & output waveform
Waveform	

$V_{in\text{-}pp}$ (the peak-to-peak voltage of input signal)	Output voltage (theoretical calculation)	Output voltage (experimental calculation)
V	V	V

A_v (theoretical calculation)	A_v (experimental calculation)

Component list

Component	Specification	Number
Op-Amp	UA741	1
Resistor	100 (1/4 W)	
	1 kΩ (1/4 W)	1
	4.7 kΩ (1/4 W)	1
Capacitor	0.1 μF	1
DC Power Supply	GW Instek GPE-3323 (Kaohsiung) UTP-3305-II (Hsinchu)	1
Signal Source (Function Generator)	Keysight EDUX1052G (Kaohsiung) GW Instek AFG-2225 (Hsinchu)	1
Oscilloscope	Keysight EDUX1052G (Kaohsiung) OSD-1102 (Hsinchu)	1
Multimeter	GW Instek GDM-532 (Kaohsiung) DHA DMM-93B+ (Hsinchu)	1

Question

1. In a practical Miller integrator circuit, a feedback resistor R_f is typically connected in parallel with the feedback capacitor C.

(a) What is the primary purpose of adding this resistor R_f?

(b) If R_f is removed (open circuit), what will happen to the output voltage V_{out} over time when the input V_{in} is zero or has a small DC offset voltage? Explain briefly in terms of capacitor charging.

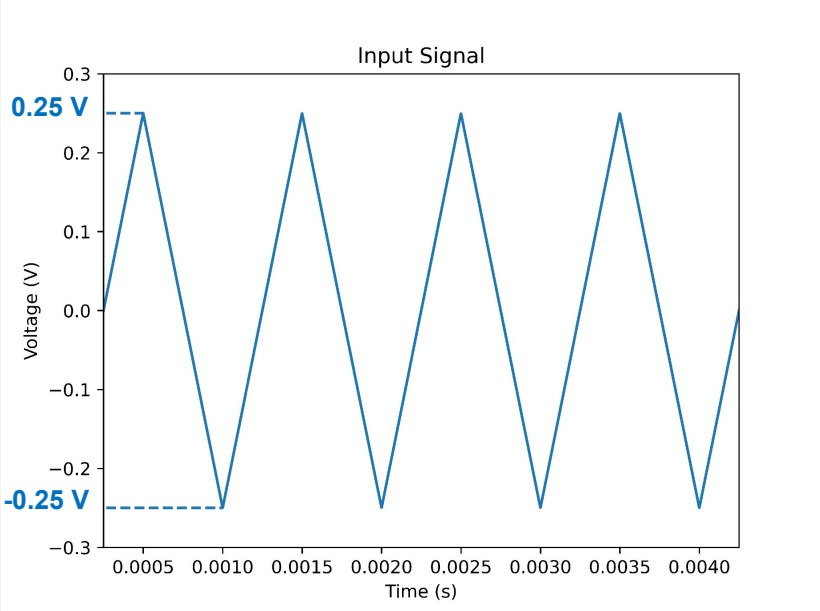
2. An ideal differentiator (with input capacitor C and feedback resistor R_f) suffers from instability and high-frequency noise amplification because its gain increases linearly with frequency. To fix this, a resistor R_s is often added in series with the input capacitor.

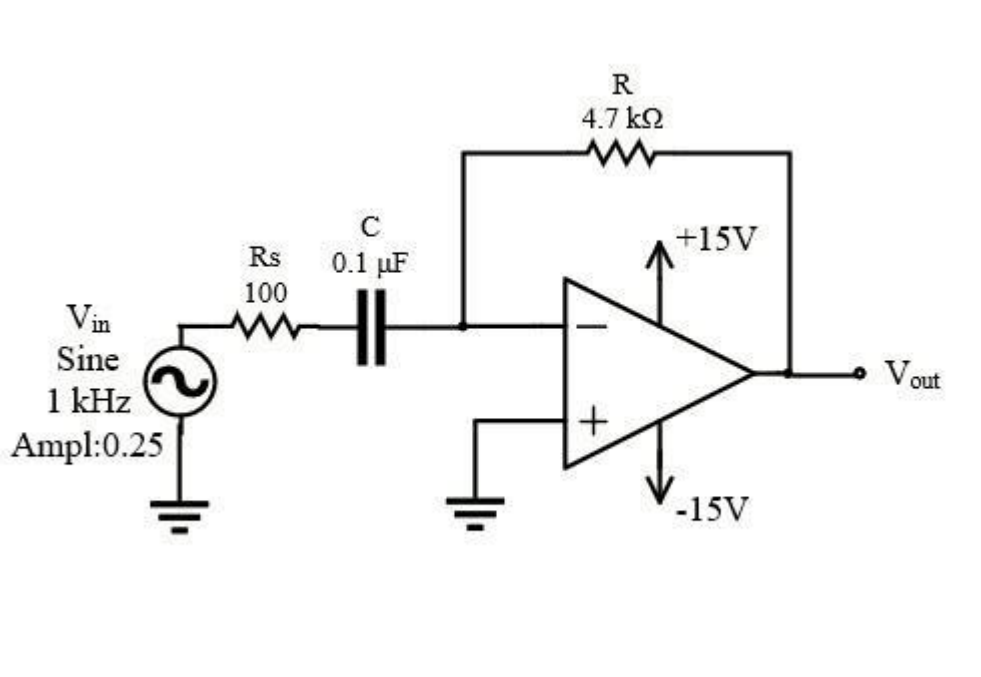
State the magnitude of the voltage gain ($|A_v| = |V_{out}/V_{in}|$) of this practical differentiator circuit at extremely high frequencies (i.e., as frequency $f \to \infty$).

Bonus

Build the op-amp differentiator circuit and input a triangular wave signal.

Record the input and output waveforms as well as the output peak-to-peak voltage V_{pp} from the oscilloscope, then compare these results with theoretical calculations and Spice simulations.





Hint: Phase delay can be ignored here.